Assignment-02

Design and Implementation of a Multi-Step Prompt Workflow for Personalized Academic Study Planning and Quiz Generation

Submitted by:

Name: DASARI RAM MOHAN

Registration No: 99230040291

Programme: GEN-AI

Module: Prompt Engineering

Table of Contents

1. Introduction
2. Objective
3. System Design
4. Methodology
4.1 Step 1: Learning Requirement Analysis
4.2 Step 2: Study Plan and Quiz Generation
4.3 Step 3: Review and Personalization
5. Results and Discussion
6. Conclusion

1. Introduction

Students often have difficulty converting a broad syllabus into a practical daily study plan. They may also spend additional time deciding what topics to revise and which questions to practice. A multi-step prompt workflow can organize this process by first understanding the learner's requirements, then creating a structured study plan and practice questions, and finally reviewing the generated material for relevance and difficulty. This assignment presents a three-stage prompt workflow for personalized academic planning.

2. Objective

The main objective is to design a controlled prompt workflow that transforms a student's syllabus, available study time, current skill level, and learning goals into a personalized study schedule and quiz set. The workflow should produce structured outputs, maintain consistency between stages, and provide a final review step before the material is presented to the student.

3. System Design

The proposed system is organized as a pipeline of three specialized stages. Stage 1 identifies the learner's requirements and converts the syllabus into structured topics. Stage 2 uses this information to create a study plan and practice questions. Stage 3 checks the generated content and adjusts it according to the learner's level and goals.

Stage 1 – Analysis: Extracts subjects, topics, learning goals, available time, and difficulty level.

Stage 2 – Generation: Creates a daily study plan, revision activities, and topic-based quiz questions.

Stage 3 – Review: Checks coverage, difficulty, clarity, and alignment with the learner's objectives.

4. Methodology

The workflow uses three connected prompt stages. Each stage receives the structured output from the previous stage. This reduces ambiguity because the later prompts do not need to reinterpret the original request from scratch.

4.1 Step 1: Learning Requirement Analysis

The first stage analyzes the student's syllabus and learning information. It identifies subjects, important topics, current proficiency, target outcome, available study hours, preferred learning style, and examination date when provided.

Prompt 1: Act as an academic learning analyst. Analyze the student's syllabus, current skill level, available study time, learning goals, and examination requirements. Extract the subjects and important topics. Group the topics by difficulty and priority. Return a structured learner profile and topic-priority summary.

4.2 Step 2: Study Plan and Quiz Generation

The second stage uses only the structured learner profile and topic-priority summary produced in Step 1. It creates a realistic schedule containing learning, revision, and practice activities. It also generates questions for the most important topics so that the student can measure understanding.

Prompt 2: Using only the learner profile and topic-priority summary from Step 1, create a practical study plan. Distribute topics across the available study sessions, include revision and practice time, and generate a short quiz for each major topic. Keep the questions appropriate to the student's current level and learning goal.

4.3 Step 3: Review and Personalization

The third stage works as a quality-control step. It checks whether the plan covers the important topics, whether the workload is realistic, and whether the quiz questions match the requested difficulty. It then produces a corrected final version.

Prompt 3: Act as a strict academic reviewer. Examine the generated study plan and quiz for topic coverage, time feasibility, difficulty balance, clarity, repetition, and alignment with the learner's objective. First identify problems, then provide the corrected final study plan and quiz.

5. Results and Discussion

The multi-step academic planning workflow provides a systematic method for turning a syllabus into actionable study activities. The analysis stage establishes a structured understanding of the student's needs before generation begins. The generation stage converts those requirements into a timetable and practice material. The review stage adds a checkpoint that can identify overloaded schedules, missing topics, unsuitable question difficulty, or unclear instructions.

Because each stage has a specific responsibility, the workflow is easier to inspect and modify than a single large prompt. For example, the analysis prompt can be improved without changing the quiz-generation prompt. Similarly, the review stage can enforce consistency before the final output is delivered to the student.

6. Conclusion

The proposed multi-step prompt workflow demonstrates how prompt engineering can be applied to personalized academic planning. Instead of asking an AI system to perform every task in one instruction, the workflow separates requirement analysis, content generation, and final validation into connected stages. This approach provides clearer checkpoints and makes the final study plan and quiz more structured, relevant, and controllable. The same workflow can be adapted for different subjects, examination patterns, and student skill levels.